



30-Minute Quick Setup

1. Project Structure (1 minute)

```
mkdir webapp-project && cd webapp-project
mkdir data/ src/ model/ demo/
```

2. Python Setup (5 minutes)

```
# One command setup
python -m venv venv && source venv/bin/activate && pip install fastapi uvicorn transformers
```

3. Docker Quick Install (10 minutes)

Windows (simplest):

1. Download Docker Desktop from [docker.com](https://www.docker.com) ^[1] ^[2]
2. Run installer, enable WSL 2 ^[3] ^[1]
3. Restart when prompted

Linux (fastest):

```
# Single command install
curl -fsSL https://get.docker.com -o get-docker.sh && sudo sh get-docker.sh
```

4. OpenResty via Docker (2 minutes)

Skip local installation - use Docker:

```
# Pull and run OpenResty container
docker run -d -p 80:80 --name openresty openresty/openresty:alpine
```

5. Tomcat + Sample WAR (10 minutes)

Quick Tomcat via Docker:

```
# Download sample WAR
wget https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war

# Run Tomcat with WAR mounted
docker run -d -p 8080:8080 --name tomcat -v $(pwd)/sample.war:/usr/local/tomcat/webapps/s
```

Alternative - Manual Tomcat:

```
# Quick download & setup
wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.80/bin/apache-tomcat-9.0.80.tar.gz
tar -xzf apache-tomcat-9.0.80.tar.gz
cd apache-tomcat-9.0.80
cp ../sample.war webapps/
./bin/startup.sh
```

6. Quick Verification (2 minutes)

```
# Test everything
python --version          # Should show Python
docker --version          # Should show Docker
curl http://localhost     # Should show OpenResty
curl http://localhost:8080/sample/ # Should show Tomcat app
pip list | grep fastapi    # Should show installed packages
```

Docker Compose Alternative (Ultra Quick - 5 minutes)

Save this as `docker-compose.yml`:^[4]

```
version: '3'
services:
  openresty:
    image: openresty/openresty:alpine
    ports:
      - "80:80"
  tomcat:
    image: tomcat:9
    ports:
      - "8080:8080"
    volumes:
      - ../sample.war:/usr/local/tomcat/webapps/sample.war
```

Then run:

```
wget https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war
docker-compose up -d
```

Total time: 25-30 minutes max^[5] ^[6]

The Docker approach is fastest since it skips complex local installations. You'll have working OpenResty, Tomcat with a sample WAR, and all Python dependencies ready for development and traffic generation.



1. <https://docs.docker.com/desktop/setup/install/windows-install/>
2. <https://docs.docker.com/get-started/introduction/get-docker-desktop/>
3. <https://stackoverflow.com/questions/75727062/how-to-install-docker-desktop-on-a-different-drive-location-on-windows>
4. <https://www.youtube.com/watch?v=IDxna2YevqQ>
5. <https://kinsta.com/blog/install-docker-ubuntu/>
6. <https://hub.docker.com/r/openresty/openresty>
7. <https://www.youtube.com/watch?v=-hdhnT5aQpw>
8. <https://www.freecodecamp.org/news/how-to-setup-virtual-environments-in-python/>
9. <https://doc.openresty.com/en/xray/on-premise/docker-compose-deployment/>
10. <https://stackoverflow.com/questions/48174599/is-there-a-single-line-way-to-run-a-command-in-a-python-venv>
11. <https://stackoverflow.com/questions/49379346/docker-openresty-flexible-configuration-how-to>
12. <https://realpython.com/python-virtual-environments-a-primer/>
13. <https://forums.docker.com/t/create-docker-container-with-both-mysql-and-tomcat/142359>
14. https://www.w3schools.com/python/python_virtualenv.asp
15. <https://docs.docker.com/desktop/setup/install/linux/ubuntu/>
16. <https://discuss.python.org/t/how-to-run-a-project-with-a-specific-virtual-environment/54209>
17. https://www.reddit.com/r/docker/comments/1azqzmv/whats_the_most_basic_way_to_install_docker_on/
18. <https://doc.openresty.com.cn/en/edge/edge-installation-upgrade/node-container-installation/>
19. <https://docs.python.org/3/library/venv.html>
20. <https://openresty.org/en/installation.html>